CAREERPATH

Career Assessment and Student Profile Management System

Phase 1 – Project Report

Project Type	MERN Stack Web Application
Academic Program	B.Tech – Artificial Intelligence and Machine Learning
Developer	Aishwarya NR
Project Status	Phase 1 – Completed
Database	MongoDB / MongoDB Atlas
Backend	Node.js + Express.js
Frontend	React.js + Vite

Submitted as a Phase 1 MERN Stack internship/project deliverable

DECLARATION

I hereby declare that the project titled **“CareerPath – Career Assessment and Student Profile Management System”** is developed as a Phase 1 MERN Stack project. The application is intended to provide students with a structured platform for managing their profiles, completing a career assessment, analysing interests and skills, and viewing career-domain recommendations.

The work presented in this report describes the implemented application, its architecture, major modules, technology stack, security approach, testing activities, screenshots, results, limitations, and proposed future enhancements.

Developer	Aishwarya NR
Program	B.Tech – Artificial Intelligence and Machine Learning
Project Phase	Phase 1 – Completed

ABSTRACT

CareerPath is a MERN Stack web application designed to help students explore suitable career domains using a structured assessment of interests, skills, academic background, strengths, and career goals. The system provides authentication, student profile management, an interactive questionnaire, interest and skill analysis, career recommendation, and a dashboard that brings the student's information together in one place.

The application follows a client–server architecture. The React frontend provides the user interface, React Router supports navigation and protected pages, Axios communicates with REST APIs, and the Node.js/Express backend handles business logic and authentication. MongoDB, accessed through Mongoose, stores user and profile-related information. JWT is used for authentication, while bcrypt is used for password hashing.

Phase 1 establishes the core foundation of the platform. The implemented recommendation domains include Software Development, Data Science, UI/UX Design, Digital Marketing, Cybersecurity, and Business Analytics. Future versions can extend the system with AI-powered recommendations, skill-gap analysis, learning paths, course and job recommendations, resume analysis, and an administrator dashboard.

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1. INTRODUCTION

Choosing a career is an important decision for students, but many students have difficulty connecting their interests, strengths, academic background, and skills with realistic career domains. CareerPath addresses this problem by providing a single digital platform where students can create an account, build a profile, complete an assessment, and receive a clear set of career-domain recommendations.

The system is developed using the MERN stack. This technology choice provides a modern JavaScript-based development environment, a responsive frontend, RESTful backend services, and a flexible document database. The Phase 1 implementation focuses on the core user journey from registration through assessment and recommendation.

2. PROBLEM STATEMENT

Students often depend on informal advice, scattered online resources, or personal assumptions when selecting a career path. Such approaches may not systematically consider a student's skills, interests, strengths, academic background, and goals.

The problem is therefore to develop an easy-to-use web application that collects relevant student information, provides a structured career assessment, analyses responses, and presents suitable career domains in a simple dashboard.

3. OBJECTIVES

- Develop a secure student registration and login system.
- Allow students to create and update an academic and personal career profile.
- Provide an interactive career assessment questionnaire.
- Analyse assessment responses to identify interest and skill patterns.
- Generate career-domain recommendations from the assessment result.
- Provide a student dashboard for viewing progress and recommendations.
- Store application data securely using MongoDB and Mongoose.
- Implement protected routes and authentication using JWT.
- Create a responsive and user-friendly interface.

4. SCOPE OF THE PROJECT

The Phase 1 scope covers the complete core student journey: account creation, login, profile completion, assessment, analysis, recommendation, and dashboard viewing. The system is intended as a foundation for a broader career guidance platform rather than a replacement for professional career counselling.

- Student-focused web application.

- Six initial career domains: Software Development, Data Science, UI/UX Design, Digital Marketing, Cybersecurity, and Business Analytics.
- REST API based communication between frontend and backend.
- MongoDB-based persistence.
- Responsive interface suitable for desktop and smaller screens.

5. FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

Functional requirements define what the system does, including registration, login, profile management, assessment submission, analysis, recommendation, and dashboard display.

Non-functional requirements define how the system should behave, including usability, security, responsiveness, maintainability, reliability, and reasonable performance.

- Authentication: register, login, logout/session handling, and protected application areas.
- Profile: maintain academic information, interests, strengths, and related career details.
- Assessment: answer multiple-choice questions and move through the questionnaire.
- Analysis: process responses to derive career-domain scores or matches.
- Recommendation: display the strongest career match together with other suitable domains.
- Dashboard: show assessment/profile status and career information.
- Security: hash passwords and protect sensitive API routes.

6. TECHNOLOGY STACK

Frontend: React.js, JavaScript, HTML5, CSS3, React Router, Axios, Lucide React, and Vite.

Backend: Node.js, Express.js, REST API, JWT authentication, and bcrypt.

Database: MongoDB, Mongoose, and MongoDB Atlas.

Development tools: Visual Studio Code, Git, GitHub, and Postman.

Layer	Technology	Purpose
Presentation	React + Vite	Build the interactive student interface
Routing	React Router	Page navigation and protected routes
HTTP Client	Axios	Frontend-to-backend API communication
Backend	Node.js + Express	REST API and server-side logic
Authentication	JWT + bcrypt	Session/token security and password hashing
Database	MongoDB + Mongoose	Persist users and career-related data
Testing/API Tool	Postman	API request and response validation

7. SYSTEM ARCHITECTURE

CareerPath follows a three-layer client–server architecture. The React/Vite client is responsible for presentation and user interaction. The Express backend exposes REST APIs, validates requests, applies authentication middleware, and coordinates database operations. MongoDB stores persistent application data.

```
Student
|
v
React Frontend (Vite)
|
| REST API / Axios
v
Node.js + Express Backend
|
| Mongoose
v
MongoDB / MongoDB Atlas
```

8. PROJECT STRUCTURE

The project is separated into client and server applications. This separation keeps presentation logic independent from backend services and database operations.

```
CareerPath_Phase1_MERN_Project/
■■■ client/
■ ■■■ public/
■ ■■■ src/
■ ■ ■■■ components/
■ ■ ■■■ context/
■ ■ ■■■ pages/
■ ■ ■■■ services/
■ ■■■ package.json
■■■ server/
■ ■■■ models/
■ ■■■ routes/
■ ■■■ controllers/
■ ■■■ middleware/
■ ■■■ server.js
■ ■■■ package.json
■■■ Phase 1 Documentation/
■ ■■■ Project Report.pdf
■ ■■■ Screenshots/
```

```
■■■ .gitignore
■■■ README.md
```

9. APPLICATION WORKFLOW

The primary workflow begins when a student creates an account. After authentication, the student completes the profile, answers the career assessment, and submits responses. The system analyses the response pattern and displays suitable career domains on the recommendation screen and dashboard.

Registration → Login → Student Profile → Career Assessment → Interest & Skill Analysis → Career Recommendation → Student Dashboard

10. MAJOR MODULES

10.1 User Authentication: Provides account registration and login. Passwords are protected using bcrypt, and JWT is used to authorize protected resources.

10.2 Student Profile: Captures academic information, interests, strengths, and other details required for career analysis.

10.3 Career Assessment: Presents structured questions with multiple response options and tracks the student's answers.

10.4 Interest and Skill Analysis: Interprets assessment responses and associates them with career-domain characteristics.

10.5 Career Recommendation: Presents the strongest match and additional suitable career domains with percentage-style match indicators.

10.6 Student Dashboard: Provides a central view of profile/assessment status and the student's career journey.

11. CAREER RECOMMENDATION APPROACH

The Phase 1 recommendation feature uses a structured assessment model. Each question provides response choices, and the selected responses contribute to the student's interest/skill profile. The resulting profile is compared with predefined characteristics for the supported career domains.

The recommendation interface presents a primary career match and additional matches. The screenshot evidence shows Business Analytics as the top match in the demonstrated user flow, followed by Data Science, UI/UX Design, Cybersecurity, Digital Marketing, and Software Development with relative match percentages.

This approach is intentionally extensible. Future versions can replace or augment the rule/scoring model with a machine-learning or AI recommendation engine after collecting sufficient labelled data.

12. DATABASE DESIGN

MongoDB is used as the persistence layer and Mongoose provides schema modelling and validation. The exact implementation can be extended as the application grows; the core data model should maintain separation between authentication data, student profile information, assessment responses, and recommendation results.

A practical logical design for the implemented modules is shown below.

Collection / Entity	Representative Data	Purpose
Users	name, email, passwordHash	Authentication and account identity
Profiles	education, specialization, interests, strengths, goals	Student career profile
Assessments	questions, selected answers, score/result	Assessment attempt and analysis data
Recommendations	career domains, match scores	Persist or present recommendation results

13. API AND BACKEND DESIGN

The backend follows a REST API approach. Routes receive HTTP requests from the React client, controllers or route handlers process application logic, middleware validates authentication where required, and Mongoose performs database operations.

- Authentication endpoints for registration and login.
- Profile endpoints for retrieving and updating student information.
- Assessment endpoints for obtaining/submitting assessment data.
- Recommendation/dashboard endpoints for retrieving student-specific results.
- Middleware for validating JWT-based access to protected resources.

14. SECURITY AND AUTHENTICATION

Security is a core part of the application because the system stores student account and profile information. Authentication uses JSON Web Tokens, while passwords are hashed with bcrypt rather than stored as plain text.

- JWT-based authentication for protected resources.
- bcrypt password hashing.
- Protected frontend routes for authenticated pages.
- Backend authentication middleware.
- Sensitive configuration stored through environment variables.
- The MongoDB connection string and JWT secret should never be committed to GitHub.

15. USER INTERFACE AND SCREENSHOTS

The following screenshots were provided with the project materials and are included as visual evidence of the Phase 1 implementation.

Login Page

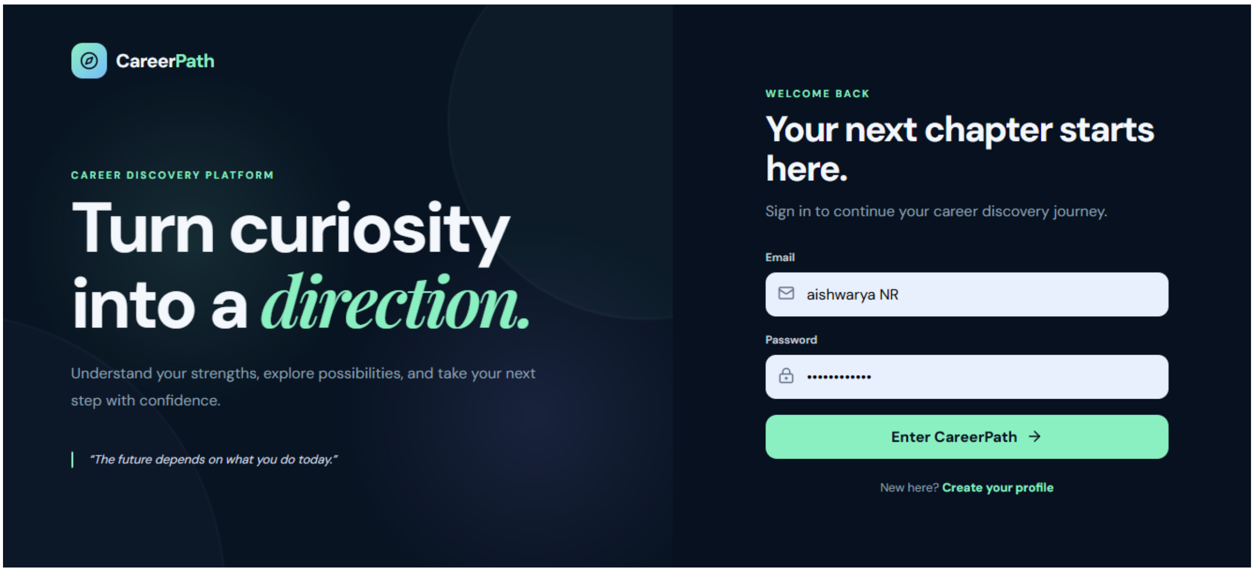

Figure Login Page: CareerPath application screen.

Registration Page

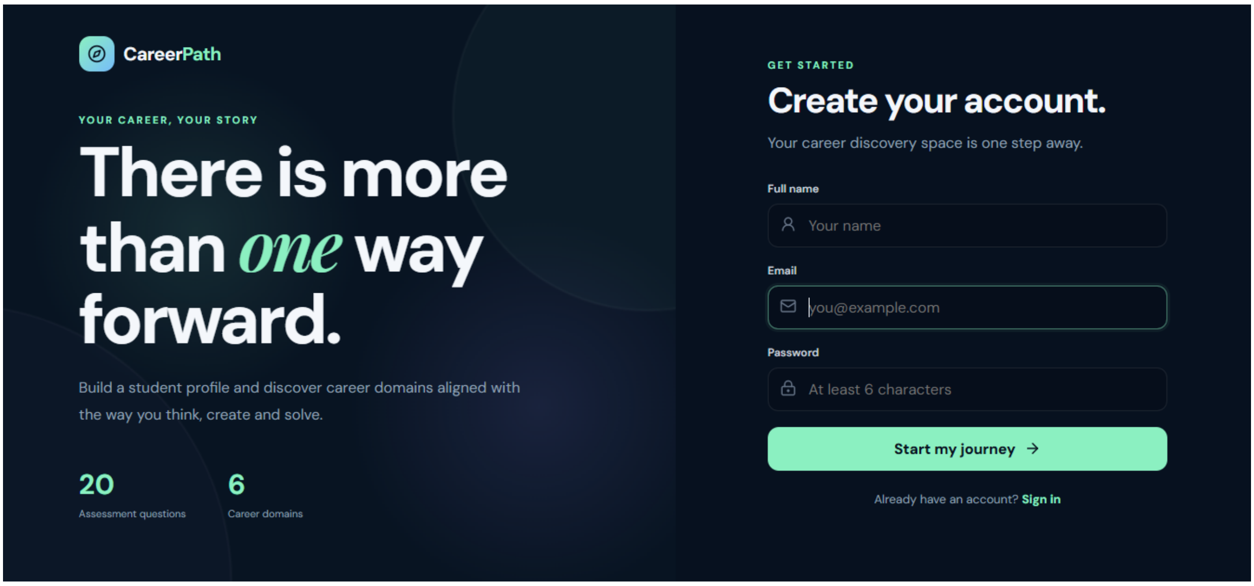

Figure Registration Page: CareerPath application screen.

Student Dashboard

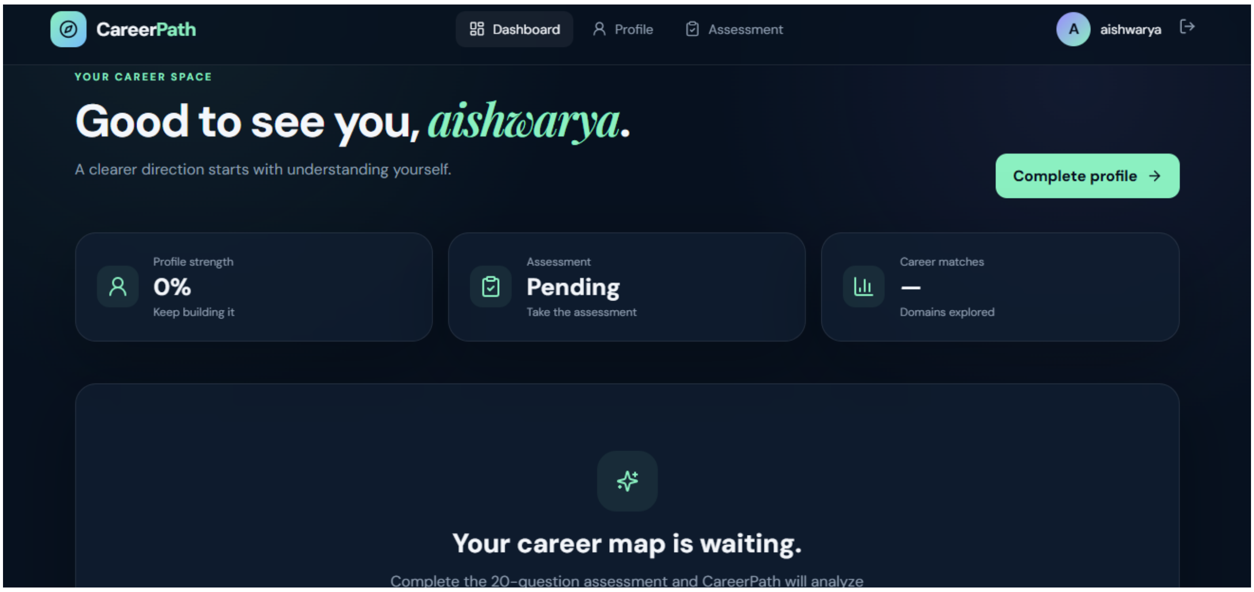

Figure Student Dashboard: CareerPath application screen.

Student Profile

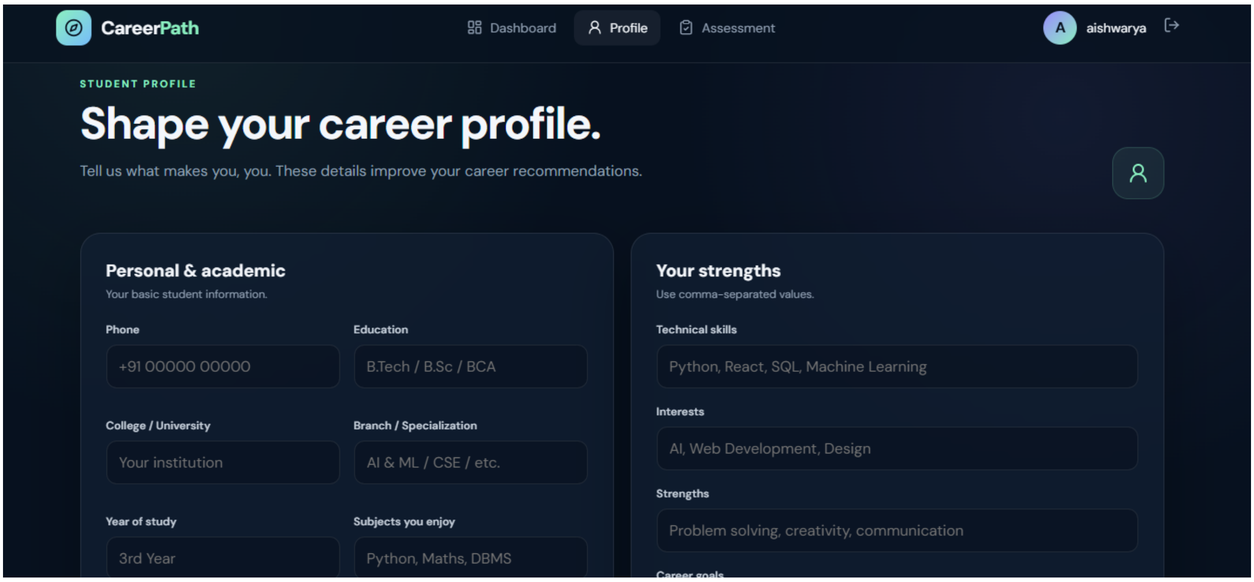

Figure Student Profile: CareerPath application screen.

Career Assessment

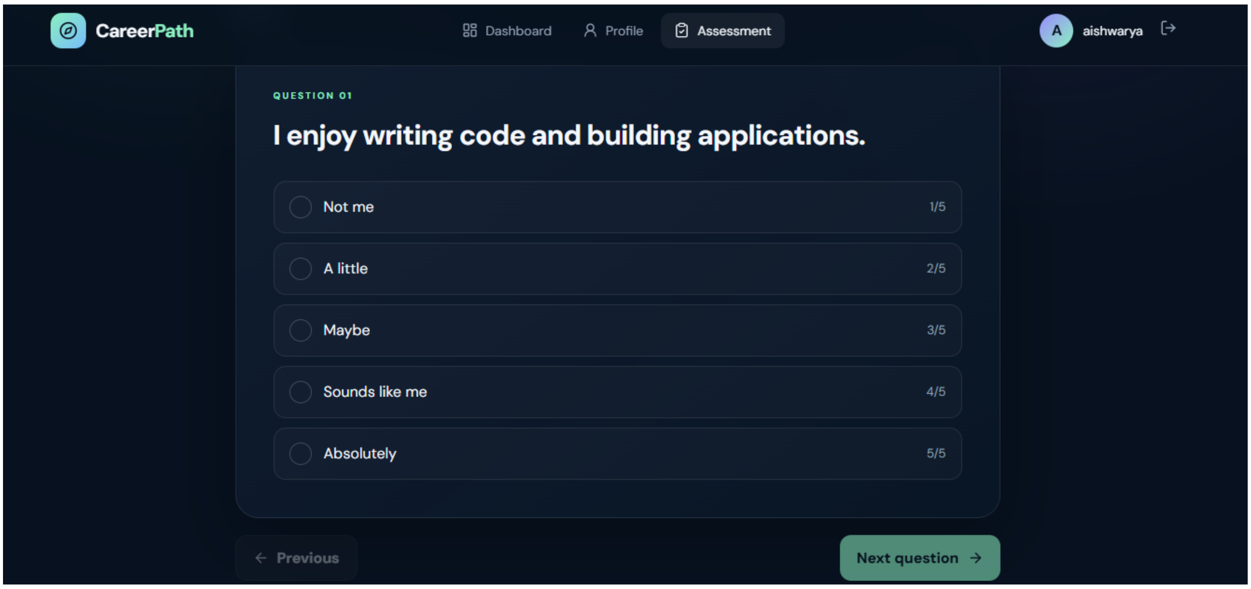

Figure Career Assessment: CareerPath application screen.

Career Recommendation

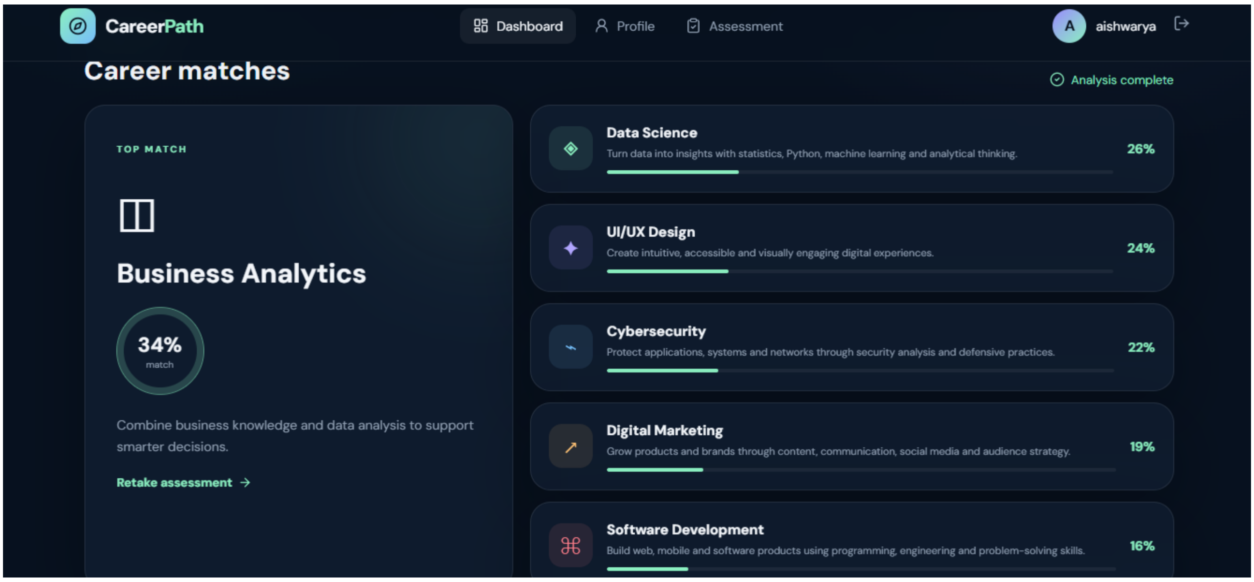

Figure Career Recommendation: CareerPath application screen.

16. TESTING AND VALIDATION

Testing was planned around the main user journey and the application's major modules. The purpose was to confirm that users can move through the system from account creation to recommendation without breaking the protected workflow.

Test Area	Expected Result	Status
Registration	New student account can be created with valid data	Passed
Login	Valid credentials allow access to the application	Passed
Protected pages	Unauthenticated users cannot access protected areas	Passed
Profile	Student information can be displayed/updated	Passed
Assessment	Questions and response options are displayed correctly	Passed

Analysis	Assessment responses produce a career-domain result	Passed
Recommendation	Primary and additional career matches are displayed	Passed
Dashboard	Student status and career information are shown	Passed
Database	Application data can be persisted through MongoDB	Passed
Responsive UI	Core screens remain usable on different screen sizes	Passed

17. RESULTS AND DISCUSSION

The completed Phase 1 application establishes a working career-guidance foundation. The supplied screenshots demonstrate the major user-facing stages: authentication, profile creation, assessment, dashboard, and recommendation.

The recommendation screen provides a clear visual ranking of career domains, while the profile and assessment screens provide the inputs required to make the recommendation process meaningful. The dashboard acts as a central progress view, helping the student understand whether the profile and assessment steps are complete.

The modular MERN architecture also makes the application suitable for future expansion. New questions, career domains, scoring rules, learning resources, and administrative functions can be added without replacing the entire system.

18. LIMITATIONS

- The current recommendation model is based on a predefined assessment/scoring approach and is not yet a trained AI model.
- Career recommendations should be treated as guidance rather than a definitive career decision.
- The Phase 1 version does not yet include a complete job-market or course recommendation engine.
- Advanced analytics and historical assessment comparison can be added in later phases.
- An administrative dashboard and content-management workflow are outside the current Phase 1 scope.

19. FUTURE ENHANCEMENTS

- AI-powered and machine-learning-based career recommendations.
- Skill-gap analysis between a student's current skills and target career.
- Personalized learning paths and course recommendations.
- Resume upload and automated resume analysis.
- Job and internship recommendations.
- Career trend and labour-market analysis.
- Admin dashboard for managing questions, domains, and recommendation rules.

- Assessment history and progress analytics.
- Email notifications and personalized career reports.
- Deployment to cloud hosting with production-grade monitoring.

20. CONCLUSION

CareerPath successfully establishes the Phase 1 foundation of a MERN Stack career assessment and student profile management system. It brings together authentication, profile management, assessment, interest and skill analysis, career recommendations, and a student dashboard in a single web application.

The project demonstrates practical use of React, Node.js, Express, MongoDB, Mongoose, JWT, bcrypt, REST APIs, and modern frontend tooling. The modular structure and secure authentication approach provide a strong base for future phases in which AI recommendations, learning paths, job matching, and advanced analytics can be introduced.

21. REFERENCES

- React.js documentation – official React documentation.
- Node.js documentation – official Node.js documentation.
- Express.js documentation – official Express documentation.
- MongoDB documentation – official MongoDB documentation.
- Mongoose documentation – official Mongoose documentation.
- JSON Web Token (JWT) documentation and standard references.
- Vite documentation – official Vite documentation.
- Postman documentation – API development and testing reference.

PHASE 1 DELIVERABLE CHECKLIST

Deliverable	Status
User Authentication	Completed
Student Profile Management	Completed
Career Assessment	Completed
Interest & Skill Analysis	Completed
Career Recommendation	Completed
Student Dashboard	Completed
Database Integration	Completed
Responsive UI	Completed

Project Status: Phase 1 – Completed

Developer: Aishwarya NR

Program: B.Tech – Artificial Intelligence and Machine Learning